Husnu S. Narman

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ACADEMIC EXPERIENCE

Associate Professor: Marshall University — **Aug 2023 - Present**
Department of Computer Sciences and Electrical Engineering — *Huntington WV, USA*

Assistant Professor: Marshall University — **Aug 2017 - May 2023**
Department of Computer Sciences and Electrical Engineering — *Huntington WV, USA*

Instructor: Marshall University — **Jan 2017 - May 2017**
Department of Computer Sciences and Electrical Engineering — *Huntington WV, USA*

Postdoctoral Fellow: Clemson University — **May 2016 - Dec 2016**
Department of Electrical and Computer Engineering — *Clemson SC, USA*

EDUCATION

Ph.D. in Computer Science — **May 2016**
University of Oklahoma — *Norman OK, USA*

M.S. in Computer Science — **May 2011**
University of Texas at San Antonio — *San Antonio TX, USA*

B.S. in Mathematics — **Aug 2006**
Abant Izzet Baysal University — *Bolu, Turkey*

HONORS

Weisberg Service Award
College of Engineering and Computer Sciences, Marshall University — *2021 - 2022*

Distinguished Artists and Scholars Junior Category Award
Marshall University — *2020 - 2021*

Weisberg Academy of Distinguished Teachers Award
College of Engineering and Computer Sciences, Marshall University — *2020 - 2021*

Outstanding Service Award
IEEE/ACM IoTDI — *2021*

Nomination for Pickens Queen Teaching Award
Marshall University — *2018, 2021*

Outstanding Ph.D. Student in Computer Science
University of Oklahoma — *2015 - 2016*

Study Abroad Fellowship for Higher Education
Turkish Ministry of National Education — *2007 - 2016*

Graduation as High Honor Student
Abant Izzet Baysal University — *2006*

RESEARCH INTEREST

Advanced Learning Technologies: *Investigate usage of high-tech products such as Virtual and Augmented Reality based applications with their effects in learning for K-12 and higher education.*

Data Mining: *Using sentiment analysis tools to understand the user behaviors for various applications like shopping habits and chain effects.*

Resource Allocation: *Determine scheduling models and allocations policies in Cloud and Fog Computing, IoT, and Networks with their applicable cases such as Vehicular Networks and Crowdsourcing.*

Smart Health: *Develop and investigate IoT products and mobile applications with machine learning algorithms to make the health system more accessible and productive by helping doctors and patients.*

R&D GROUP ATTAINMENTS

Fellowship & Scholarship

- Josh Maddy (Undergraduate): NASA Undergraduate Research Fellowship for Spring 2023; *Project: The Metaphysical Exhibition An Exploration of Technology, the Arts, and Sciences: Expansion*
- Neil Loftus (Undergraduate): NASA Undergraduate Research Fellowship for Spring 2023; *Project: Detecting Birds with Real Time Image Processing for Drone Safety: Expansion*
- Neil Loftus (Undergraduate): NASA Undergraduate Research Fellowship 2022 - 2023; *Project: Detecting Birds with Real Time Image Processing for Drone Safety*
- Neil Loftus (Undergraduate): Marshall University Creative Discovery and Research Award for Summer 2022; *Project: The Cybersecurity Packet Control Simulator: The Effect of Visual Learning Tools on Retention of Information in Computer Science*
- Josh Maddy (Undergraduate): Marshall University Creative Discovery and Research Award for Summer 2022; *Project: The Metaphysical Exhibition an Exploration of Technology, the Arts, and Sciences*
- Josh Maddy (Undergraduate): Marshall University SURE Summer Research Fellowship 2022; *Project: Augmented Reality as an Aid for Physics Concepts*
- Neil Loftus (Undergraduate): Marshall University SURE Summer Research Fellowship 2022; *Project: The Cybersecurity Packet Control Simulator*
- Eric Dillion (Undergraduate): Marshall University Creative Discovery and Research Award for Spring 2022; *Project: Automatic Feedback System to Teach Cybersecurity Principles*
- Eric Shoemaker (Undergraduate): Marshall University Creative Discovery and Research Award for Summer 2021; *Project: Crowdsourcing based Community Infrastructure Management Application*
- Eric Shoemaker (Undergraduate): NASA Undergraduate Research Fellowship for Spring 2021; *Project: Community Infrastructure Management Application*
- Jarred Carter (Undergraduate): NASA Undergraduate Research Fellowship 2020 - 2021; *Project: Simulation for Trade-off Model of Fog-Cloud Computing for Space Network*
- William Coleman (Undergraduate): Marshall University SURE Summer Research Fellowship 2020; *Project: Enhancing STEM Education with Augmented Reality*
- James Farley (Undergraduate): NASA Undergraduate Research Fellowship for Spring 2020; *Project: Emotion Classification of Users in Social Media*
- Alex Canfield, Cameron Berry, Jeremy Giese, Logan Carpenter (Undergraduates): Marshall University Research Scholar Award for Spring 2019; *Project: Data Structure with Augmented Reality*
- James Farley (Undergraduate): Marshall University Creative Discovery and Research Award for Fall 2019; *Project: Emotion Classification of Users based on the Comments and Emojis in Social Media*
- Jarred Carter (Undergraduate): NASA Undergraduate Research Fellowship 2019 - 2020; *Project: Trade-off Model of Fog-Cloud Computing for Space Network*
- Caleb Kesler (Undergraduate): Marshall University Creative Discovery and Research Award for Summer 2019; *Project: Development of Story-Assisted Platform for Early Childhood for Coding*
- Alymbek Damir Uulu (Undergraduate): Marshall University SURE Summer Research Fellowship 2019; *Project: Profile Analysis on Cryptocurrency Investors and Social Engineering on Their Prices*
- Jared Lee Lewis (Undergraduate): Marshall University Creative Discovery and Research Award for Fall 2018; *Project: Automated IP Reputation Analyzer System with Machine Learning*
- Geanina Tambaliuc (Undergraduate): Marshall University SURE Summer Research Fellowship 2018; *Project: Automated IP Reputation Analyzer System*

- Alex Kacinari, Chris Murphy, Derek M Staley (Undergraduates): Marshall University Research Scholar Award for Spring 2018; *Project: Suturing Technique Simulation*
- Charlie Murphy, Michael B Branard, Steven D. Gunnels (Undergraduates): Marshall University Research Scholar Award for Spring 2018; *Project: Embedded Storybook Game*

Internship/Development

- Hwapyeong Song (Graduate): WV Department of Education Internship and Paving Academy 2020
- Neil Loftus (High School): Paving Academy 2020
- Geanina Tambaliuc (Undergraduate): WV Department of Education Internship 2019
- Anh Nguyen (Undergraduate): WV Department of Education Internship 2019
- Jake Gressang (Undergraduate): WV Department of Education Internship 2019
- Kuo Chi Fang (Graduate): WV Department of Education Internship 2018
- Ibrahim Hussein Mwinyi (Graduate): WV Department of Education Internship 2018

Publications

- High School Students: Isabella Schrader, Laina Karim, Neil Loftus, Sawyer Slack.
- Undergraduate Students: Alex Canfield, Alymbek Damir Uulu, Anh Nguyen, Cameron Berry, Cameron Green, Craig Carpenter II, Eric M. Dillon, Eric Shoemaker, Geanina Tambaliuc, Greg Weed, James Farley, Jared Lee Lewis, Jarred Carter, Jeremy Giese, John Cook, Josh Maddy, Logan Carpenter, Neil Loftus, Thomas D. Wills, William Coleman.
- Graduate Students: Govind Yatnalkar, Ibrahim Hussein Mwinyi, Kanimozhi Kalaichelavan, Kuo Chi Fang, Sreehari Sreenath.

TAUGHT COURSES AND TEACHING EFFECTIVENESS

Q1	I believe that I learned in this class.
Q2	The course was well organized.
Q3	This course challenged me intellectually.
Q4	I have become more competent in this area because of this course.
Q5	The objectives of the course were well explained.
Q6	The instructor followed his/her syllabus.
Q7	The instructor gave clear explanations to clarify concepts.
Q8	The instructor was supportive in academic situations.
Q9	The instructor showed enthusiasm when teaching.
Q10	The instructor informed students of their progress.
Q11	The instructor's use of examples helped to get points across in class.
Q12	The instructor adequately explained the grading scale.
Q13	The instructor treated me fairly.
Q14	The instructor was enthusiastic about the course material.
Q15	The instructor encouraged students to ask questions.
Q16	The instructor provided me with an effective array of challenges.
Q17	The instructor carefully answered questions raised by students.
Q18	The instructor treated students with respect.
Q19	The instructor presented material in a clear manner.
Q20	The instructor used class time well.
Q21	The instructor seemed genuinely interested in wanting me to learn.
Q22	I would recommend this instructor to other students.

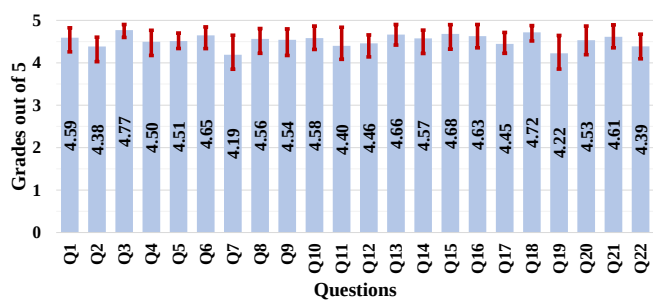

Figure 1: Each question (left table) is asked to more than 429 undergraduate and graduate students to grade from 1 to 5 in course evaluations. The figure bars show the average grades with minimum and maximum values for each question (the average is out of 5).

Table 1 shows the taught courses with student evaluation results for each course and semester. Undergraduate courses are from 1xx to 4xx, and graduate courses are from 5xx and above. Mostly, a high number of students responded the student evaluation. Figure 1 also shows the course evaluations for each question.

Table 1: *Enrollment* shows the number of registered students, *R.Rate* shows the percentage of students who answered the course evaluation; and scores show the mean score out of 4 for each course and semester.

Term	**Course**	**E.ment**	**R.Rate**	**C.Score**	**S.Score**
Spring 2017	CS 210 Data Structures and Algorithms	12	75%	3.64	3.49
	CS 215 Adv. Data Structures and Algorithms	17	65%	3.22	
	CS 410 Database Engineering	23	83%	3.28	
	CS 510 Database Systems	29	76%	3.75	
	IS 623 Database Management	13	85%	3.72	3.72
Fall 2017	CS 210 Data Structures and Algorithms	20	90%	3.73	3.72
	CS 651 Cloud Computing	16	94%	3.70	
Spring 2018	CS 210 Data Structures and Algorithms	13	85%	3.72	3.77
	CS 510 Database Systems	10	80%	3.83	
	IS 623 Database Management	27	96%	3.77	3.77
Fall 2018	CS 210 Data Structures and Algorithms	16	88%	3.79	3.84
	CS 360 Automata and Formal Languages	21	76%	3.76	
	CS 651 Cloud Computing	18	100%	3.95	
Spring 2019	CS 120 Computer Science II (OOP-Java)	20	75%	3.64	3.63
	CS 210 Data Structures and Algorithms	14	86%	3.60	
	CS 681 Master Thesis	1	100%	3.71	
Fall 2019	CS 120 Computer Science II (OOP-Java)	18	94%	3.65	3.71
	CS 210 Data Structures and Algorithms	12	83%	3.79	
	CS 681 Master Thesis	1	100%	3.89	
Spring 2020	CS 120 Computer Science II (OOP-Java)	39	74%	3.70	3.71
	CS 210 Data Structures and Algorithms	12	92%	3.75	
	CYBR 620 Cyberwarfare	1	100%	3.89	3.89
Fall 2020	CS 210 Data Structures and Algorithms (Vir)	15	100%	3.53	3.67
	CS 210 Data Structures and Algorithms (Web)	6	67%	3.62	
	CS 602 Cloud Computing (Web)	13	92%	3.87	
Spring 2021	CS 210 Data Structures and Algorithms (Vir)	27	81%	3.44	3.54
	CS 510 Advanced Database Systems (Vir)	3	100%	3.98	
	CS 510 Advanced Database Systems (Web)	4	100%	3.75	
	CYBR 620 Cyberwarfare (Web)	7	100%	3.51	3.51
Fall 2021	CS 210 Data Structures and Algorithms	15	93%	3.68	3.66
	CS 210 Data Structures and Algorithms (Web)	14	43%	3.51	
	CS 602 Cloud Computing (Web)	5	80%	3.80	
Spring 2022	CS 210 Data Structures and Algorithms	19	89%	3.59	3.51
	CS 210 Data Structures and Algorithms (Web)	11	55%	3.20	
	CS 510 Advanced Database Systems (Web)	9	78%	3.57	
	CYBR 620 Cyberwarfare (Web)	5	80%	3.71	3.71
Fall 2022	CS 210 Data Structures and Algorithms	19	89%	3.71	3.59
	CS 210 Data Structures and Algorithms (Web)	15	40%	3.34	
	CS 602 Cloud Computing (Web)	7	86%	3.41	
Spring 2023	CS 210 Data Structures and Algorithms	15	73%	3.66	3.51
	CS 210 Data Structures and Algorithms (Web)	15	73%	3.32	
	CS 510 Advanced Database Systems (Web)	11	100%	3.50	
	CS 685 Independent Study	1	100%	4.00	

STUDENT ADVISING

Table 2 shows the number of advised students in each semester. Graduate students have not been advised between Spring 2018 and Fall 2020 due to policy changes of the Department of Computer Sciences and Electrical Engineering for advising.

Table 2: The number of students advised in each semester.

Term	Total	Undergraduate	Graduate
Spring 2017	18	9	9
Fall 2017	35	27	8
Spring 2018	27	26	1
Fall 2018	41	41	0
Spring 2019	39	39	0
Fall 2019	39	39	0
Spring 2020	27	27	0
Fall 2020	32	30	2
Spring 2021	21	18	3
Fall 2021	36	29	7
Spring 2022	36	28	8
Fall 2022	36	29	7
Spring 2023	33	24	9

SERVICES

Marshall University **Huntington WV, USA**

- Organization of Computer Science Adventure Zone Summer Camp 2017 - Present
- Elected College Representative on the Athletic Committee 2019 - Present
- CS Freshman Orientation 2018 - Present
- CS Online Course Development 2018 - Present
- CS Conference Organization Committee 2018 - Present
- Computer Science (CS) Student Leadership Program Coordinator 2019 - 2020
- CS Representative on the College Outreach and Recruitment Committee 2019 - 2020
- CS Outreach Committee Chair 2017 - 2020
- CS Faculty Candidate Interviewing Committee 2017 - 2019
- Catalog Editing and Correction for BS and MS Programs in CS 2017 - 2018
- CS Project Room Management 2017 - 2018

PROFESSIONAL ACTIVITIES

Co-Lead for Training Programs **2021 - Present**
Institute for Cyber Security (ICS), Marshall University

Editorial Board **2018 - Present**
Editorial Board: Elsevier Journal of Network and Computer Applications

Summer Camp on Robotics and Cybersecurity for K-12 **2017 - Present**
Marshall University *Huntington WV, USA*

Faculty Mentor Internship Program **2018 - 2020**
WV Department of Education *Huntington WV, USA*

Virtual Conference Working Group Lead
• 2021 ACM/IEEE Conference on Internet of Things Design and Implementation

Publicity Co-Chair
• 2017 International Conference on Networking, Architecture, and Storage • 2017 International Conference on Computer Communications and Networks

Technical Program Committee
• IEEE Global Communications Conf. • IEEE Int. Conf. on Communications • Springer Ubiquitous Networking Conf. • IEEE Wireless Communications and Networking Conf. • IEEE Int. Conf. on Communications, Network, and Satellite • IEEE 5G World Forum (WF-5G) • IEEE Int. Conf. on Internet of Things and Intelligence System • Elsevier Int. Conf. on Ambient Systems, Networks and Technologies • IEEE Int. Conf. on Fog and Edge Computing • IEEE Symp. on Signal Processing and Information Technology • IEEE Int, Conf. on Signals and Systems • IEEE/ACM Int. Symp. in Cluster, Cloud, and

Grid Computing • IEEE Middle East & North Africa Communications Conf. • IEEE Int. Conf. on Wireless Networks and Mobile Communications • IEEE TENCON

Journal Reviewer

• IEEE Journal on Selected Areas in Communications • IEEE Transactions on Mobile Computing • IEEE Transactions on Parallel and Distributed Systems • IEEE Transactions on Vehicular Technology • IEEE Transactions on Industrial Electronics • IEEE Transaction on Intelligent Transportation Systems • IEEE Transactions on Sustainable Computing, • ACM Transactions on Cyber-Physical Systems • ACM Transactions on Knowledge Discovery from Data • Elsevier Journal of Network and Computer Applications • Elsevier Future Generation Computer System • MDPI Sensors • Wiley Software: Practice and Experience

Membership

• IEEE Senior Member (2022 - Present) • IEEE Member (2014 - 2021)

VOLUNTEER ACTIVITIES

VEX IQ Robot Tournament for Middle and Elementary Schools **2019 - Present**
Marshall University *Huntington WV, USA*

Faculty adviser for IEEE Marshall University Student Chapter **2022 - Present**
Marshall University *Huntington WV, USA*

Faculty adviser for Geeks and Gadgets Club **2018 - Present**
Marshall University *Huntington WV, USA*

Coordinator and Initiator for Community Service Leadership Program **2019 - 2020**
Marshall University *Huntington WV, USA*[-5pt]

CERTIFICATION

Responsible Conduct of Research **2018**
Collaborative Institutional Training Initiative Program *Huntington WV, USA*

Behavioral & Social Science Research **2018**
Collaborative Institutional Training Initiative Program *Huntington WV, USA*

Independent Improving Your Online Course (IYOC) **2017**
Quality Matters *Huntington WV, USA*

Java SE 7 **2016**
Robert Half Technology *Clemson SC, USA*

GRANTS

External Funded and Under Review Grant Applications

[15] *Automatic Assessment and Repair for Railroads.* **Co-PI**. ERDC, 2023. Amount: $2M, **Funded**.

[14] *Establishing a Cyber Security Center for Critical Infrastructure.* **Responsible for Training Modules**. Department of Education, 2023. Amount: Total:$1,5M, Share: $750K, **Funded**.

[13] *Detecting Birds with Real Time Image Processing for Drone Safety.* **Mentor**. NASA West Virginia Undergraduate Research Fellowship, 2022. Amount: $5K, **Funded**.

[12] *Detecting Birds with Real Time Image Processing for Drone Safety: Expansion.* **Mentor**. NASA Undergraduate Affiliate Fellowship Program, 2022. Amount: $1K, **Funded**.

[11] *Experiences for Teachers on Cyber Security.* **Proposal Developer**. NSF/NSA GenCyber Teacher Summer Camp, 2022. Amount: $140K, **Funded**.

[10] *The Metaphysical Exhibition an Exploration of Technology, the Arts, and Sciences.* **Mentor**. NASA Undergraduate Affiliate Fellowship Program, 2022. Amount: $1K, **Funded**.

[9] *Crowdsourcing Infrastructure Management System.* **Mentor**. NASA Undergraduate Affiliate Fellowship Program, 2021. Amount: $1K, **Funded**.

[8] *Community Infrastructure Management Application.* **Mentor**. NASA Undergraduate Affiliate Fellowship Program, 2020. Amount: $1K, **Funded**.

[7] *Development of the Pavement Preservation and Rehabilitation Academy.* **Co-PI**. Wirtgen Group - John Deere Company, 2020. Amount: $40K, **Funded**.

[6] *Trade-off Model of Fog-Cloud Computing for Space Network.* **Mentor**. NASA West Virginia Undergraduate Research Fellowship, 2020. Amount: $5K, **Funded**.

[5] *Emotion Classification of Users in Social Media.* **Mentor**. NASA Undergraduate Affiliate Fellowship Program, 2019. Amount: $1K, **Funded**.

[4] *Experiences for Teachers on Cyber Security.* **PI**. NSF/NSA GenCyber Teacher Summer Camp, 2019. Amount: $73K, **Funded**.

[3] *Trade-off Model of Fog-Cloud Computing for Space Network.* **Mentor**. NASA West Virginia Undergraduate Research Fellowship, 2019. Amount: $5K, **Funded**.

[2] *WV Faculty Mentor Internship Program - Calendar and Special Education Applications.* **Co-PI**. West Virginia Department of Education, 2019. Amount: $85K, **Funded**.

[1] *WV Faculty Mentor Internship Program - Complain Management Application and Security of Applications.* **Co-PI**. West Virginia Department of Education, 2019. Amount: $79K, **Funded**.

Internal Funded Grant Applications

[25] *Augmented Reality as an Aid for Physics Concepts.* **Mentor**. Undergraduate Summer Research Experience (SURE), Marshall University, 2022. Amount: $4K, **Funded**.

[24] *Auto Feedback System based on Artificial Intelligence for Cybersecurity Learners.* **PI**. Faculty Summer Research, Marshall University, 2022. Amount: $2K, **Funded**.

[23] *Automatic Feedback System to Teach Cybersecurity Principles.* **Mentor**. Undergraduate Fall Creative Discovery Scholar and Undergraduate Research Scholar Awards, Marshall University, 2022. Amount: $1,750.00, **Funded**.

[22] *Marshall University Presentation Center: Developing A Campus-Wide Resource to Enhance Communication Fluency Across the Curriculum.* **Co-PI**. Hedrick Program Grant for Teaching Innovation, Marshall University, 2022. Amount: $5K, **Funded**.

[21] *The Cybersecurity Packet Control Simulator.* **Mentor**. Undergraduate Summer Research Experience (SURE), Marshall University, 2022. Amount: $4K, **Funded**.

[20] *The Cybersecurity Packet Control Simulator: The Effect of Visual Learning Tools on Retention of Information in Computer Science.* **Mentor**. Undergraduate Summer Research and Creative Discovery, Marshall University, 2022. Amount: $5K, **Funded**.

[19] *The Metaphysical Exhibition an Exploration of Technology, the Arts, and Sciences.* **Mentor**. Undergraduate Summer Research and Creative Discovery, Marshall University, 2022. Amount: $5K, **Funded**.

[18] *Crowdsourcing Infrastructure Management System.* **Mentor**. Undergraduate Summer Research and Creative Discovery, Marshall University, 2021. Amount: $5K, **Funded**.

[17] *Enhancing STEM Education with Augmented Reality.* **Mentor**. Undergraduate Summer Research Experience (SURE), Marshall University, 2020. Amount: $4K, **Funded**.

[16] *A Smart Therapy Tool for Feeding and Speech Disorder Detection.* **PI**. John Marshall University Summer Scholars Awards, Marshall University, 2019. Amount: $6.5K, **Funded**.

[15] *Artificial Intelligence and Integrated Fog-Cloud Computing based Matching Algorithm for Carpooling.* **PI**. Faculty Summer Research, Marshall University, 2019. Amount: $2K, **Funded**.

[14] *Augmented Reality based Application for Data Structure.* **Mentor**. Undergraduate Fall Research Scholar Awards, Marshall University, 2019. Amount: $250.00, **Funded**.

[13] *Development of Story-Assisted Platform for Early Childhood for Coding.* **Mentor**. Undergraduate Summer Research and Creative Discovery, Marshall University, 2019. Amount: $5K, **Funded**.

[12] *Emotion Classification of Users based on the Comments and Emojis in Social Media.* **Mentor**. Undergraduate Fall Creative Discovery Scholar and Undergraduate Research Scholar Awards, Marshall University, 2019. Amount: $1,750.00, **Funded**.

[11] *Profile Analysis on Cryptocurrency Investors and Social Engineering on their Prices.* **Mentor**. Undergraduate Summer Research Experience (SURE), Marshall University, 2019. Amount: $4K, **Funded**.

[10] *Quinlan Endowment Faculty Travel*. **PI**. Quinlan Endowment Faculty Travel, Marshall University, 2019. Amount: $500.00, **Funded**.

[9] *Student Research and Innovation Center*. **Co-PI**. Cross-disciplinary Research Facilitation Grant, Marshall University, 2019. Amount: $500.00, **Funded**.

[8] *Automated IP Reputation Analyzer System*. **Mentor**. Undergraduate Summer Research Experience (SURE), Marshall University, 2018. Amount: $4K, **Funded**.

[7] *Automated IP Reputation Analyzer System*. **Mentor**. Undergraduate Fall Creative Discovery Scholar and Undergraduate Research Scholar Awards, Marshall University, 2018. Amount: $1,750.00, **Funded**.

[6] *Game Embedded Storybook*. **Mentor**. Undergraduate Spring Research Scholar Awards, Marshall University, 2018. Amount: $175.00, **Funded**.

[5] *Placement of Electric Vehicle Charging Sections and Traffic Management*. **PI**. Faculty Summer Research, Marshall University, 2018. Amount: $2K, **Funded**.

[4] *Quinlan Endowment Faculty Travel*. **PI**. Quinlan Endowment Faculty Travel, Marshall University, 2018. Amount: $400.00, **Funded**.

[3] *Suturing Technique Simulation*. **Mentor**. Undergraduate Spring Research Scholar Awards, Marshall University, 2018. Amount: $175.00, **Funded**.

[2] *Research Initiation and Developemnt*. **PI**. Marshall University, 2017. Amount: $100,000.00, **Funded**.

[1] *Self-dynamic Data Center Management to Optimize Search Speed for Media Type Files*. **PI**. Faculty Summer Research, Marshall University, 2017. Amount: $2,000.00, **Funded**.

PUBLICATIONS

Patents and Intellectual Property

[1] *Oral Therapy Tool, System, and Related Methods*. Patent Appl. Serial No. 62/947,264, Regular, United States. (Submitted: November 2019, Application: December 2019, Updated: October 2022, Allowed: January 2023, Granted: August 2023 – 11,712,366).

Books

[1] **Husnu S. Narman** and Mohammed Atiquzzaman. *Carrier Assignment and Packet Scheduling in LTE-A and Wi-Fi*. Dissertation as a Book. LAP LAMBERT Academic Publishing, May 2016, p. 160. ISBN: 9783659891977. URL: `https://www.amazon.com/Carrier-Assignment-Packet-Scheduling-Wi-Fi/dp/3659891975`.

Journals

[14] Neil Loftus* and **Husnu S. Narman**. "Use of Machine Learning in Interactive Cybersecurity and Network Education". In: *Sensors* 23.6 (2023). ISSN: 1424-8220.

[13] Amrit Pal, Abishi Chowdhury, Satakshi, **Husnu S. Narman**, Arkabandhu Chowdhury, and Manish Kumar. "Random Partition based Adaptive Distributed Kernelized SVM for Big Data". In: *IEEE Access* (2022).

[12] Arnob Paul, Md. Hasanul Islam, Md. Shohrab Hossain, and **Husnu S. Narman**. "A novel zone walking protection for secure DNS Server". In: *International Journal of Interdisciplinary Telecommunications and Networking* (2022).

[11] **Husnu S. Narman**, Haroon Malik, and Govind Yatnalkar^{+}. "An Enhanced Ride Sharing Model Based on Human Characteristics, Machine Learning Recommender System, and User Threshold Time". In: *Springer Journal of Ambient Intelligence and Humanized Computing* (2021). **Invited**.

[10] Jinwei Liu, Haiying Shen, Hongmei Chi, **Husnu S. Narman**, Yongyi Yang, Long Cheng, and Wingyan Chung. "A Low-Cost Multi-Failure Resilient Replication Scheme with Data Correlation for High Data Availability in Cloud Storage". In: *IEEE/ACM Transaction on Networking* (2020).

[9] Abishi Chowdhury, Shital A. Raut, and **Husnu S. Narman**. "DA-DRLS: Drift adaptive deep reinforcement learning based scheduling for IoT resource management". In: *Journal of Network and Computer Applications* 138 (May 2019), pp. 51–65.

[8] Ankur Sarker, Haiying Shen, M. Rahman, M. Chowdhury, K. Dey, F. Li, Y. Wang, and **Husnu S. Narman**. "A Review of Sensing and Communication, Human Factors, and Controller Aspects for Information-Aware Connected and Automated Vehicles". In: *IEEE Transactions on Intelligent Transportation Systems* (March 2019).

[7] Kuo-Chi Fang+, **Husnu S. Narman**, Ibrahim Hussein Mwinyi+, and Wook-Sung Yoo. "PPHA-Popularity Prediction Based High Data Availability for Multimedia Data Center". In: *International Journal of Interdisciplinary Telecommunications and Networking* 11.1 (January 2019), pp. 17–29.

[6] Jinwei Liu, Haiying Shen, **Husnu S. Narman**, Z. Lin, and Z. Li. "Popularity-aware Multi-failure Resilient and Cost-effective Replication for High Data Durability in Cloud Storage". In: *IEEE Transactions on Parallel and Distributed Systems* (October 2018).

[5] Jinwei Liu, Haiying Shen, **Husnu S. Narman**, Wingyan Chung, and Zongfang Lin. "A Survey of Mobile Crowdsensing Techniques: A Critical Component for The Internet of Things". In: *ACM Transactions on Cyber-Physical Systems* 2.3 (June 2018).

[4] Jinwei Liu, Haiying Shen, L. Yu, **Husnu S. Narman**, J. Zhai, J. O. Hallstrom, and Y. He. "Characterizing Data Deliverability of Greedy Routing in Wireless Sensor Networks". In: *IEEE Transactions on Mobile Computing* 17.3 (March 2018), pp. 543–559.

[3] **Husnu S. Narman**, Md.Shohrab Hossain, Mohammed Atiquzzaman, and Haiying Shen. "Scheduling Internet of Things Applications in Cloud Computing". In: *Annals of Telecommunications* (February 2017).

[2] **Husnu S. Narman**, Mohammed Atiquzzaman, Mehdi Rahmani-andebili, and Haiying Shen. "Joint and Selective Component Carrier Assignment in LTE-A". In: *Computer Networks* (September 2016).

[1] **Husnu S. Narman**, Md.Shohrab Hossain, and Mohammed Atiquzzaman. "Management and Analysis of Multi Class Traffic in Single and Multi-band Systems". In: *Wireless Personal Communications* 83 (July 2015).

Conferences

[44] Eric Shoemaker+, Haroon Malik, **Husnu S. Narman**, and Jamil Chaudri. "Explaining the Unseen: Leveraging XAI to Enhance the Trustworthiness of Black-Box Models in Performance Testing". In: *International Conference on Future Networks and Communications (FNC)*. Halifax, Nova Scotia, August 2023.

[43] Shawon Kumar Saha, Rafidul Islam Sarker, Md Shohrab Hossain, and **Husnu S. Narman**. "Android Malware Detection Using Incremental Learning Approach". In: *IEEE International Black Sea Conference on Communications and Networking (IEEE BLACKSEACOM)*. Istanbul, Turkey, July 2023.

[42] Josh Maddy*, Eric M. Dillon*, and **Husnu S. Narman**. "Adapting Cybersecurity Teacher Training Camp to Virtual Learning". In: *IEEE Integrated STEM Education Conference (IEEE ISEC)*. **Best Paper Award**. Laurel, Maryland, March 2023.

[41] Josh Maddy* and **Husnu S. Narman**. "Virtual Reality Museum Application for the Arts". In: *IEEE Integrated STEM Education Conference (IEEE ISEC)*. Laurel, Maryland, March 2023.

[40] Shahriar Hassan, Md. Asif Muztaba1, Shohrab Hossain, and **Husnu S. Narman**. "A Hybrid Encryption Technique based on DNA Cryptography and Steganography". In: *IEEE Information Technology, Electronics and Mobile Communication Conference (IEEE IEMCON)*. Virtual, October 2022.

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